

Low-level acoustic feature perception is modulated by speaker emotion and feature type

Morning Meeting | By: Jax Skye, PhD

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What is prosody?

- Verbal communication conveys linguistic information (what) and paralinguistic information (how)
 - Paralinguistic examples: rhythm, pitch
- **Prosody** – modulating paralinguistic features to convey things such as emotional state
 - Affective / emotional prosody
- Example: “I’m going to the mall this afternoon.”

Why do we care about prosody?

- Prosody is socially relevant
- Involves building a model of the other person's mind
- Prosody production and perception differences seen in:
 - Autism Spectrum Disorder (ASD) (Rosenblau et al., 2017)
 - Schizophrenia (Jahshan et al., 2013)
 - Major depressive disorder (Koch et al., 2018)
 - Individuals with brain lesions (Ross, 1981, Adolphs et al., 2002, Patel et al., 2018)

Relationship between low-level acoustic features and emotional prosody

Acoustic Predictors and General Description of Emotion Categories According to [Banse and Scherer \(1996\)](#)

Emotion category	Dimensions				General description
	Pitch	Temporal	Loudness	Timbre	
Hot anger	X			X	High and bright voice with limited pitch fluctuations
Panic fear	X				High-pitched voice with limited fluctuations
Anxiety	X		X		Quiet voice in the middle pitch range with limited pitch fluctuations
Desperation	X	X		X	High and bright voice with limited pitch fluctuations and a slow speech rate
Sadness			X	X	Quiet and thin voice
Elation	X			X	High-pitched voice with some fluctuations
Boredom	X	X	X		Low and quiet voice with slow speech rate
Shame			X		Quiet voice
Pride	X				Low-pitched voice
Contempt	X				Low-pitched voice with some pitch fluctuations

(Banse and Scherer, 1996)

Study Goals

- Does emotional context modulate psychoacoustic perception and vary with psychopathology traits?
- **Aim 1**: Identify differences in pitch and loudness perception across three emotional states of the speaker: happy, angry, and neutral.
- **Aim 2**: Explore the relationship between pitch and loudness sensitivity and socially relevant psychopathology traits in a nonclinical population.

Measures of Psychopathology Traits

- Autism Quotient (AQ) (Baron-Cohen, 2000)
 - Self-report of autistic traits in adults
- Social Responsiveness Scale – II (SRS) (Constantino & Gruber, 2012)
 - Self-report questionnaire to identify social impairment associated with ASDs and quantifies its severity
- Liebowitz Social Anxiety Scale (LSAS) (Liebowitz, 1987)
 - Self-report questionnaire measuring fear and avoidance in social situations

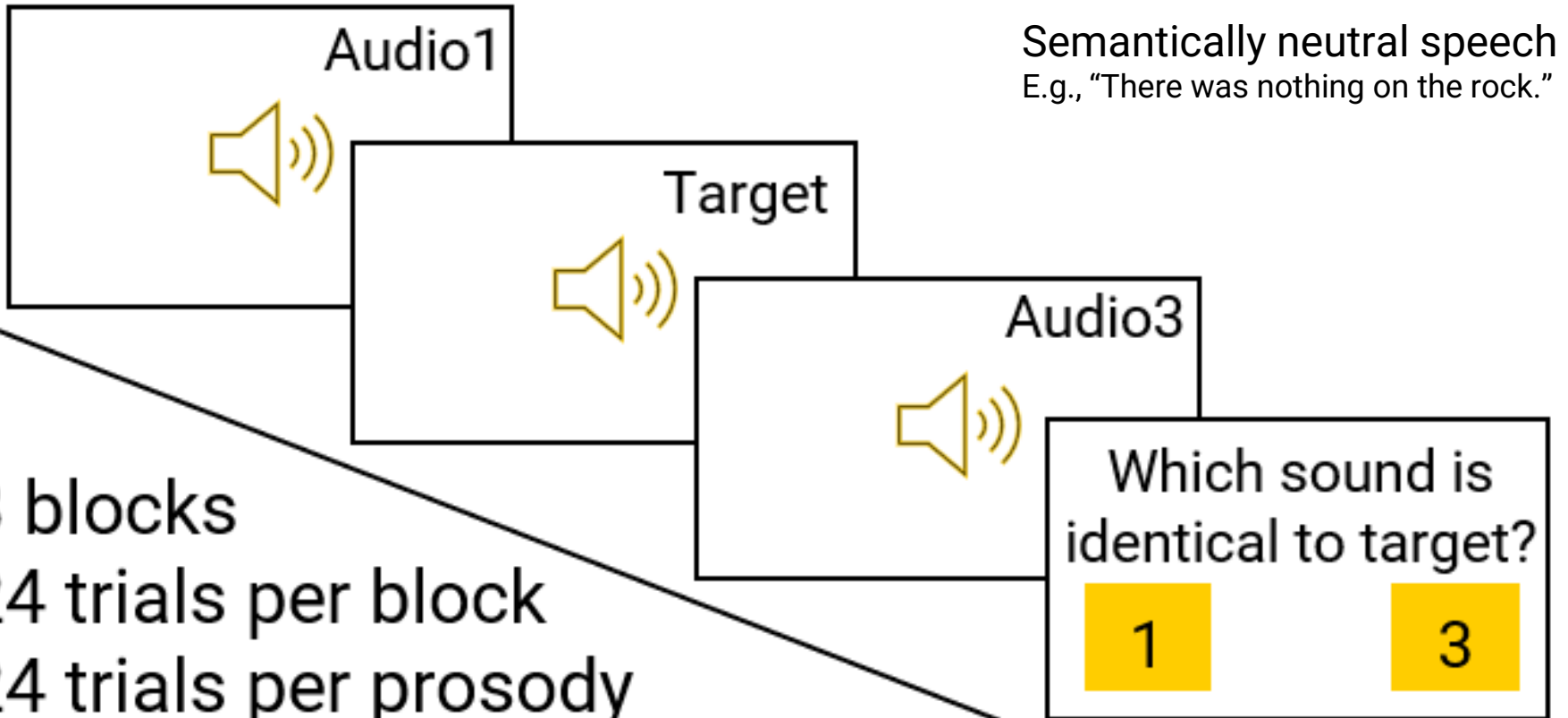
Inclusion Criteria

- At least 18
- Native English speakers
- Normal hearing
- No pre-existing psychiatric or neurological disease

Sample Demographics

	Normal	Quilted	Shuffled
N	50	53	59
Age	18.6 $\pm$ 1.06	18.75 $\pm$ 0.789	19.2 $\pm$ 1.40
Gender (W M)	35 14	37 16	45 14

Paradigm



3 blocks
24 trials per block
24 trials per prosody
(happy, angry, neutral)
trials advance on button press

Stimuli

- Stimuli from Emotional Speech Database and Emotional Voices Database
- 2 – 4 seconds in duration
- Loudness and pitch altered by 3.5%

Example of what a trial sounds like

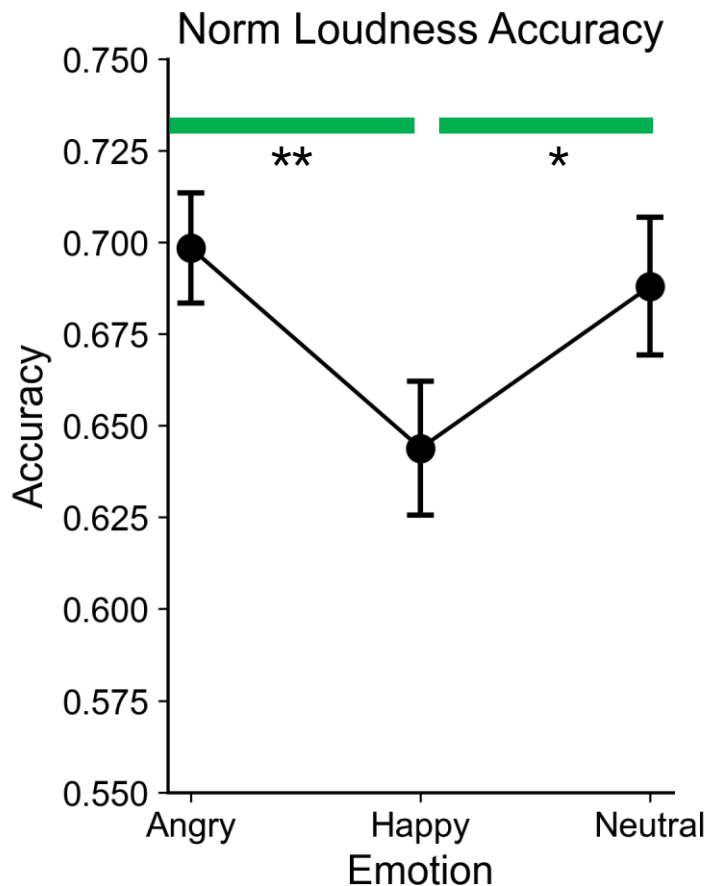


Results

Aim 1

* $p < .05$
** $p < .01$
*** $p < .001$

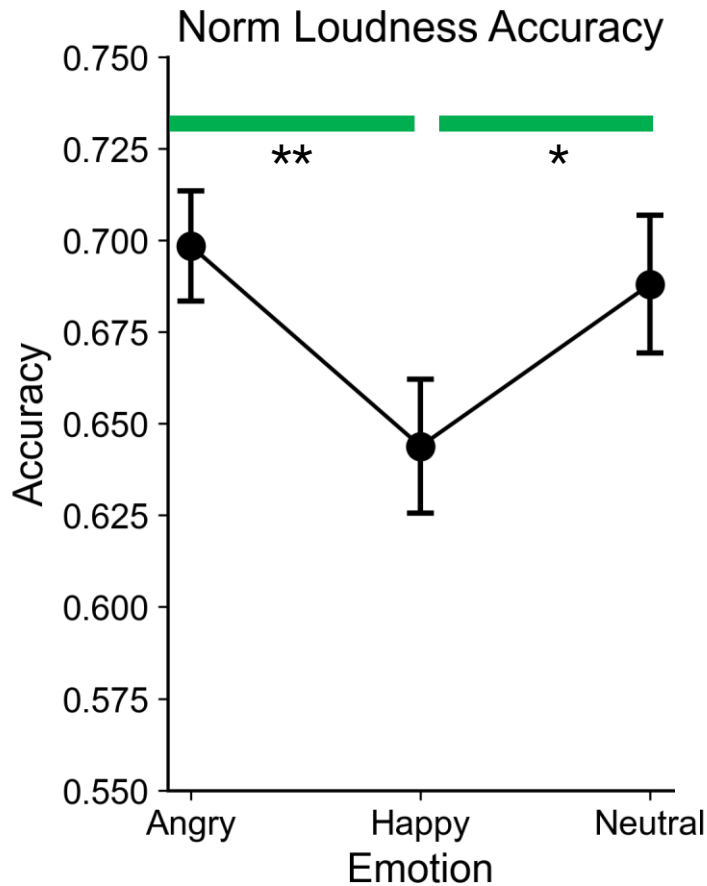
Perception of loudness modulation



- Accuracy varied across emotions ($F(2,98) = 5.674, p = .005, \eta^2_p = .104$)
- Loudness modulation for **happy prosody was less accurately identified** compared to angry ($p = 0.003, d = 0.44$) and neutral ($p = 0.025, d = 0.36$)

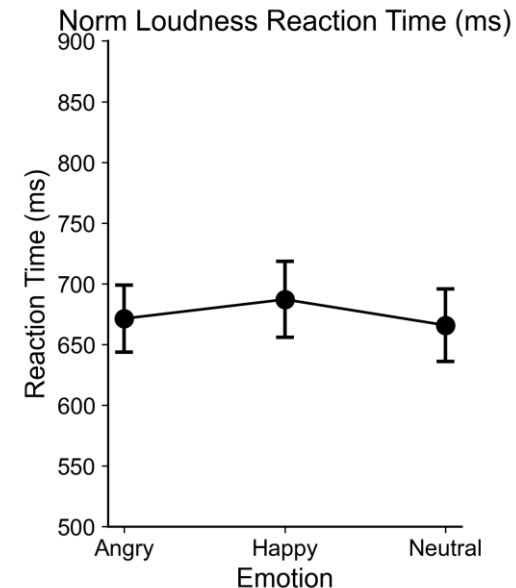
* $p < .05$
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Perception of loudness modulation



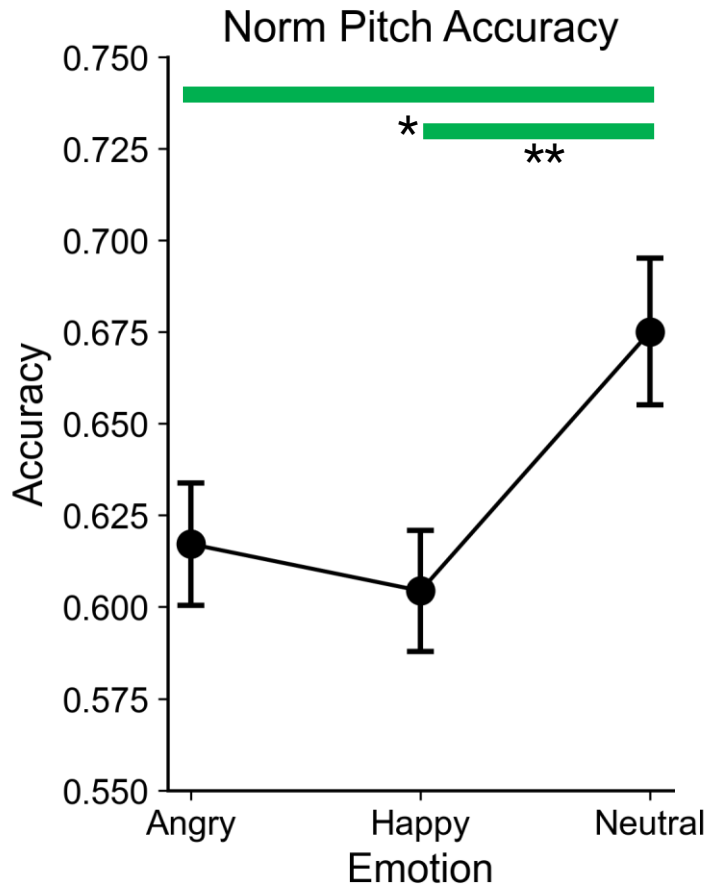
- Accuracy varied across emotions ($F(2,98) = 5.674, p = .005, \eta^2_p = .104$)
- Loudness modulation for **happy prosody was less accurately identified** compared to angry ($p = 0.003, d = 0.4$) and neutral ($p = 0.025, d = 0.36$)

No significant effects were observed for RT, $F(2,96) = .347, p = .708, \eta^2_p = .007$



* $p < .05$
** $p < .01$
*** $p < .001$

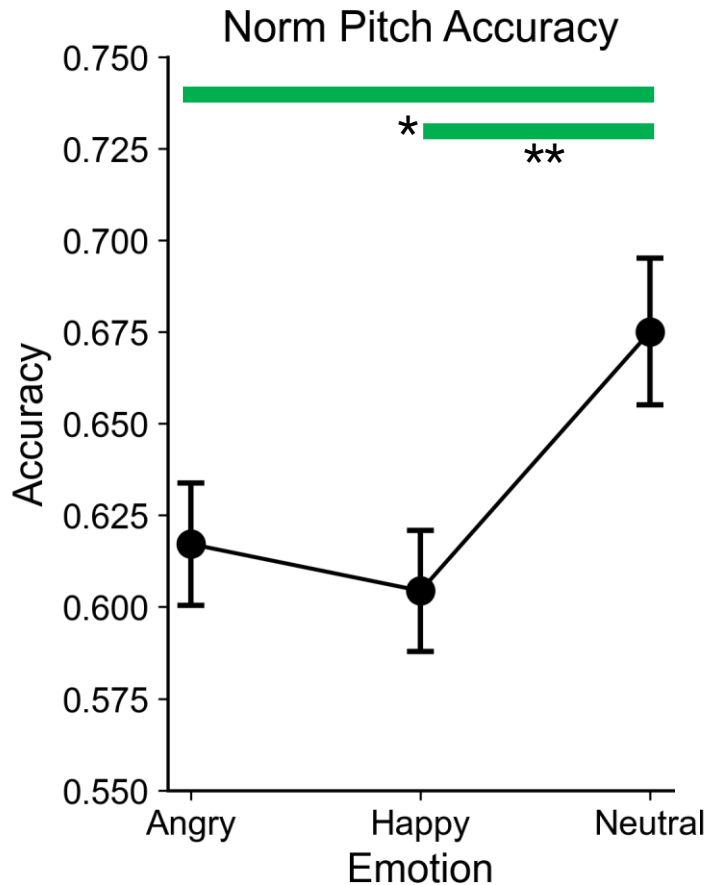
Perception of pitch modulation



- Accuracy varied across emotions ($F(2,96) = 7.429, p = .001, \eta^2p = .134$)
- Pitch modulation in **neutral was identified with higher accuracy** relative to angry ($p = 0.012, d = 0.464$) and happy ($p = 0.003, d = 0.566$)

* $p < .05$
** $p < .01$
*** $p < .001$

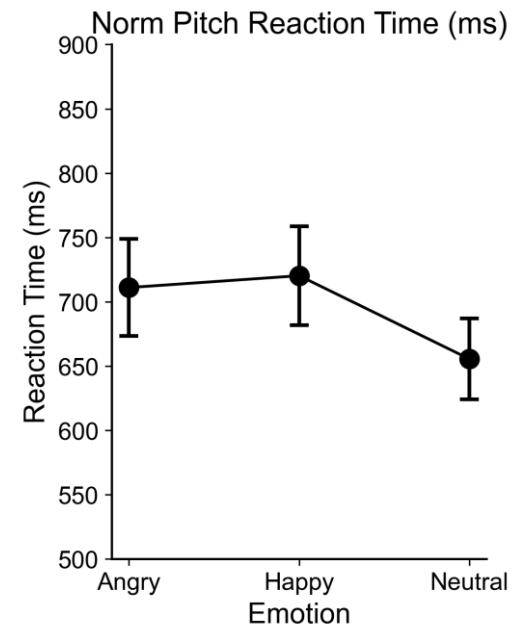
Perception of pitch modulation



- Accuracy varied across emotions ($F(2,96) = 7.429, p = .001, \eta^2_p = .134$)
- Pitch modulation was **identified with higher accuracy for neutral prosody** relative to angry ($p = 0.012, d = 0.464$) and happy ($p = 0.003, d = 0.566$)

Significant main effect observed for RT, $F(2,94) = 3.437, p = .036, \eta^2_p = .068$

Happy vs neutral trended to significance ($p = .051, d = .259$)



Aim 1 Summary

- Differences in loudness modulation were more accurately identified in the angry and neutral conditions relative to happy
- Differences in pitch modulation were more accurately identified in neutral prosody relative to emotional contexts

Results

Aim 1: Control Analyses: Normal vs Shuffled

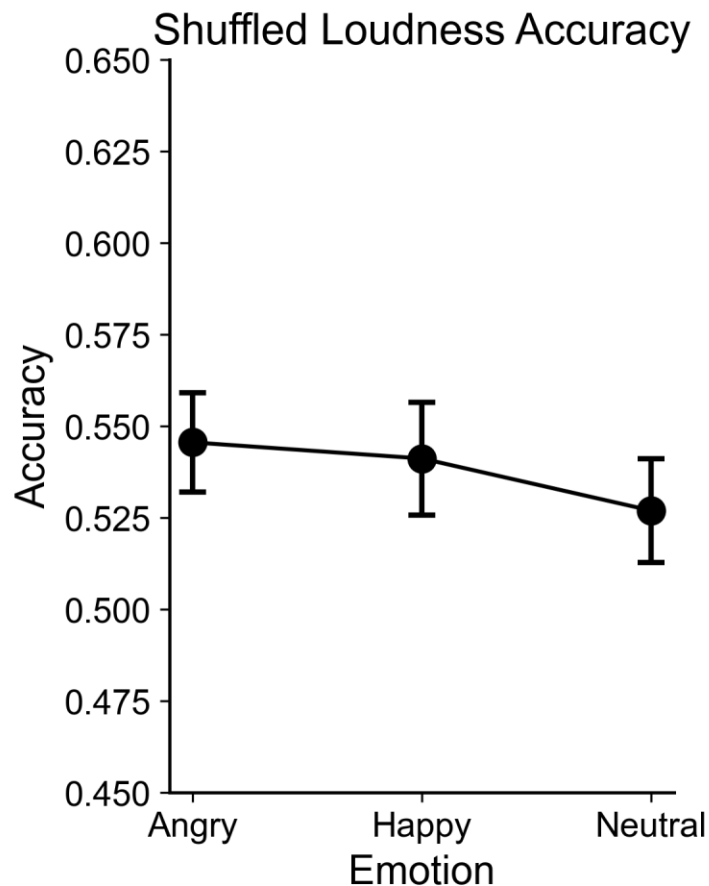


Could emotion effect be due to low level acoustic features (unrelated to emotion)?

- Shuffling:
 - Removes longer temporal structure
 - Introduces “chopped/stuttering” artifact



Loudness modulation perception (shuffled)



- Accuracy did not vary across emotions ($F(2,114) = 0.530$, $p = .590$, $\eta^2_p = .009$)

Normal vs Shuffled (Loudness)

Speech type:

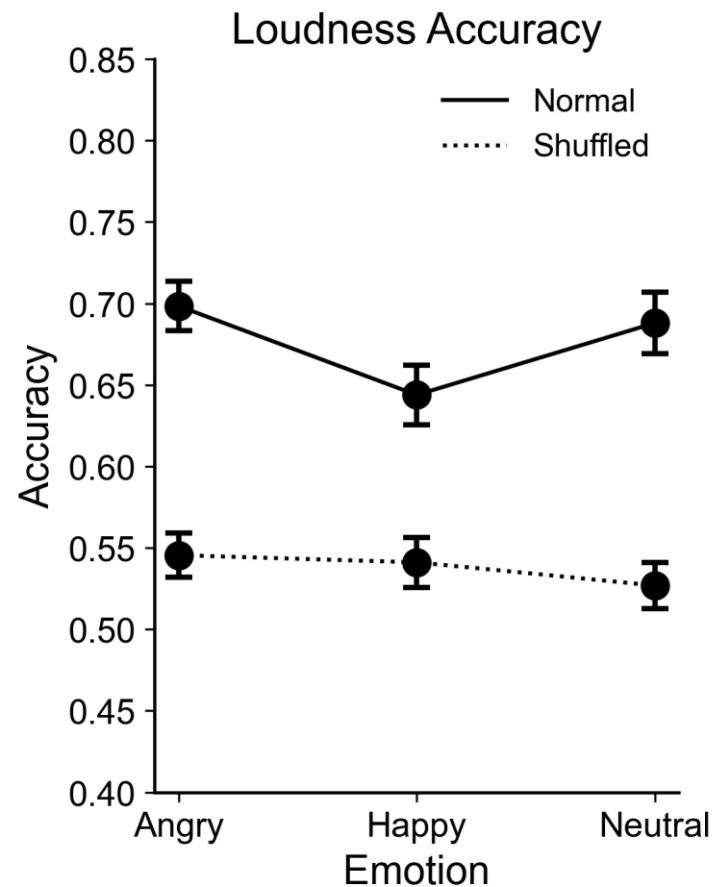
$F(1,108) = 52.762, p < .001, \eta^2_p = .328$

Emotion:

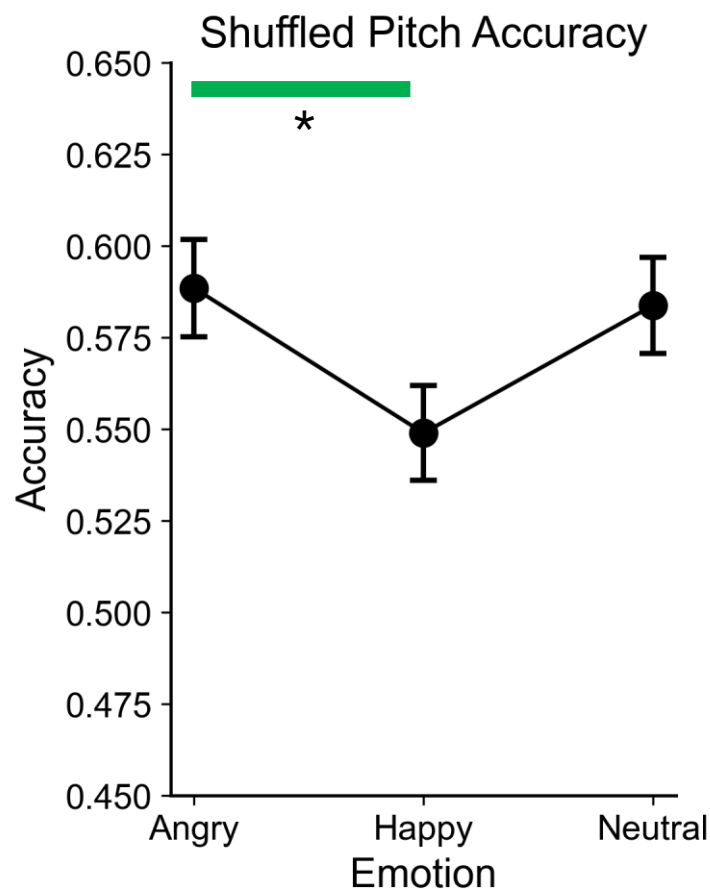
$F(2,216) = 2.451, p = .089, \eta^2_p = .022$

Interaction:

$F(2,216) = 2.963, p = .054, \eta^2_p = .027$

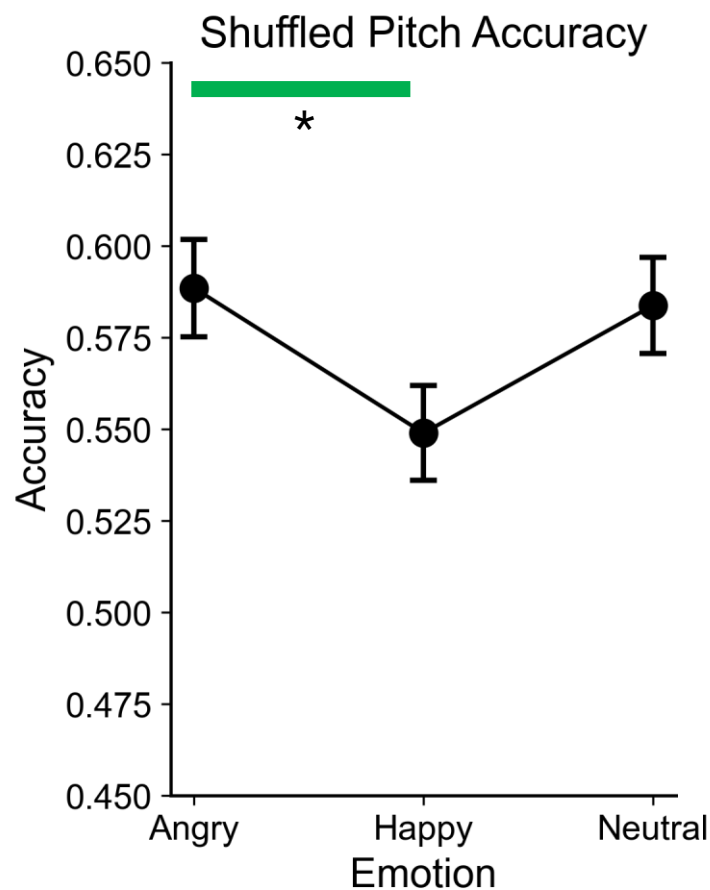


Pitch modulation perception (shuffled)

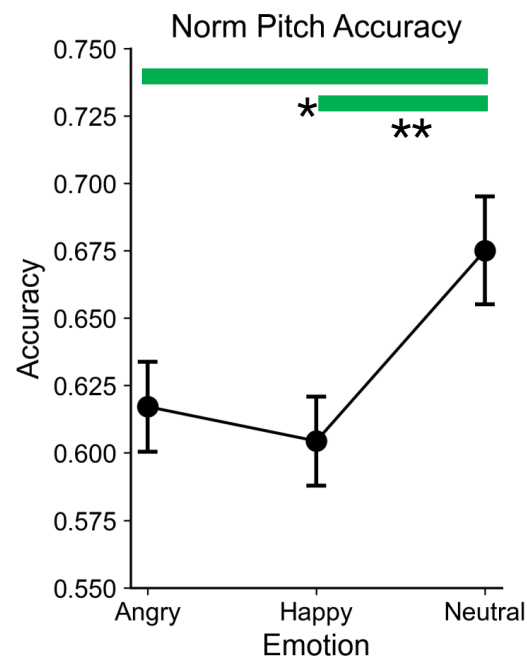


- Accuracy varied across emotions ($F(2,118) = 3.958, p = .022, \eta^2_p = .063$)
- Pitch modulation for **happy prosody was less accurately identified** compared to angry ($p = 0.031, d = 0.402$)

Pitch modulation perception (shuffled)



- Accuracy varied across emotions ($F(2,118) = 3.958, p = .022, \eta^2_p = .063$)
- Pitch modulation for **happy prosody** was **less accurately identified** compared to angry ($p = 0.031, d = 0.402$)



Normal vs Shuffled (Pitch – Accuracy)

Speech type:

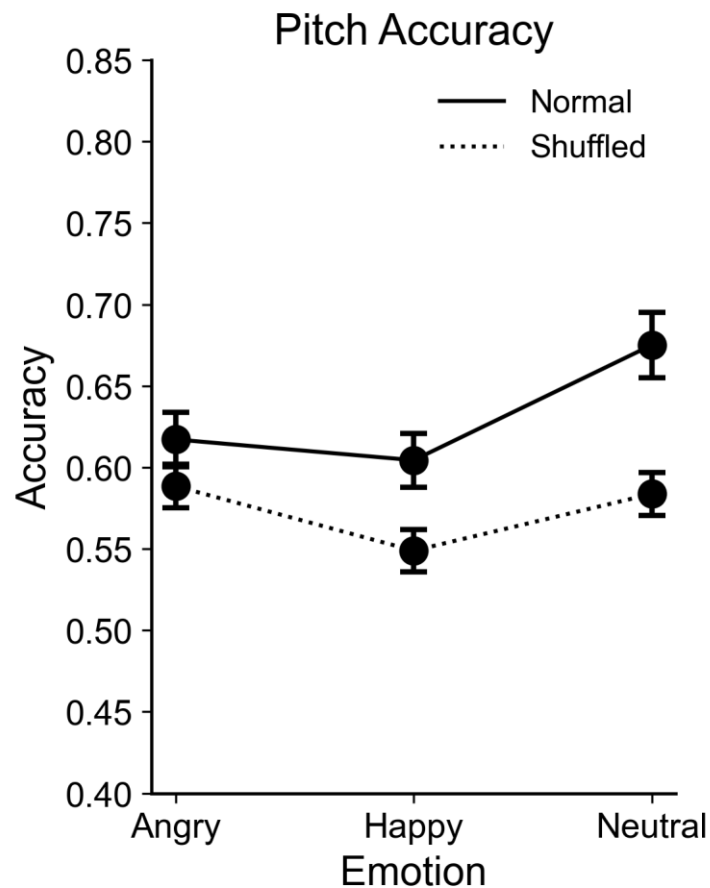
$F(1,108) = 52.762, p < .001, \eta^2_p = .125$

Emotion:

$F(2,216) = 8.940, p < .001, \eta^2_p = .076$

Interaction:

$F(2,216) = 3.714, p = .026, \eta^2_p = .033$



Results

Aim 1: Control Analyses: Normal vs Quilted

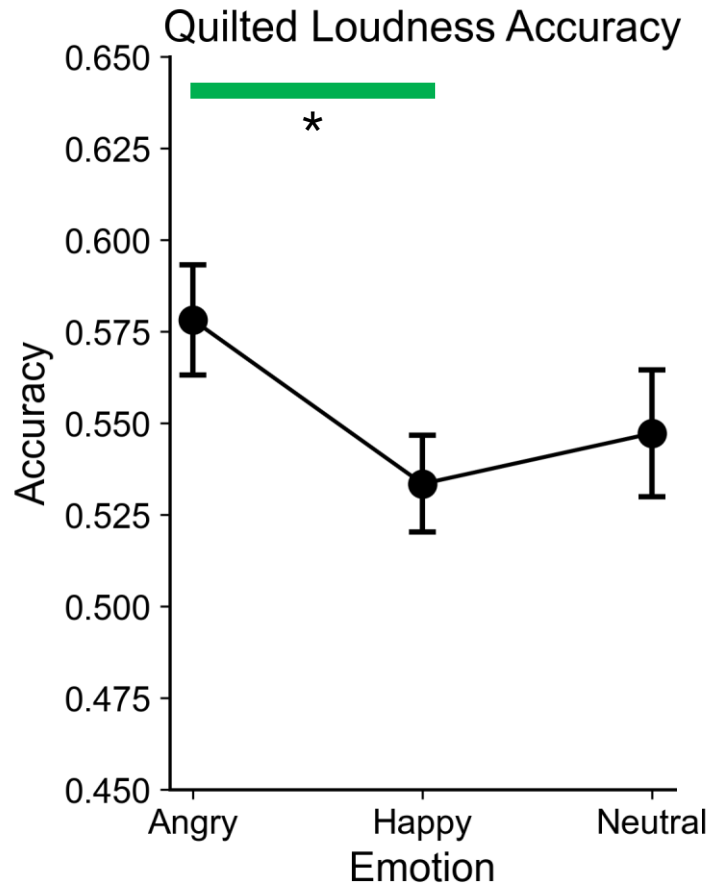


Quilted Speech

- Quilting removes feature composition of long-term structure while retaining simple spectro-temporal statistics
- Removes prosodic content



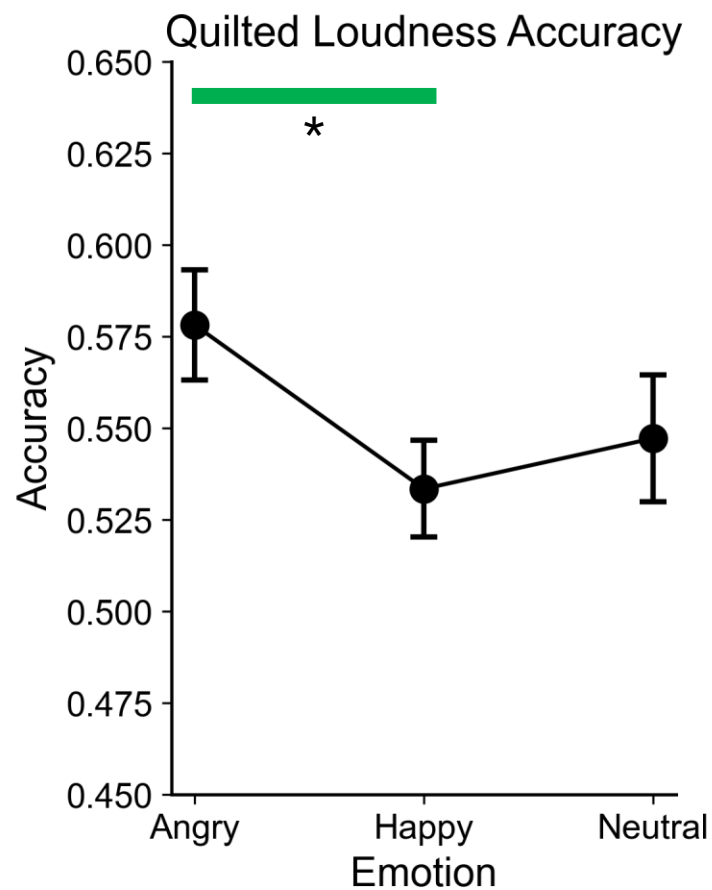
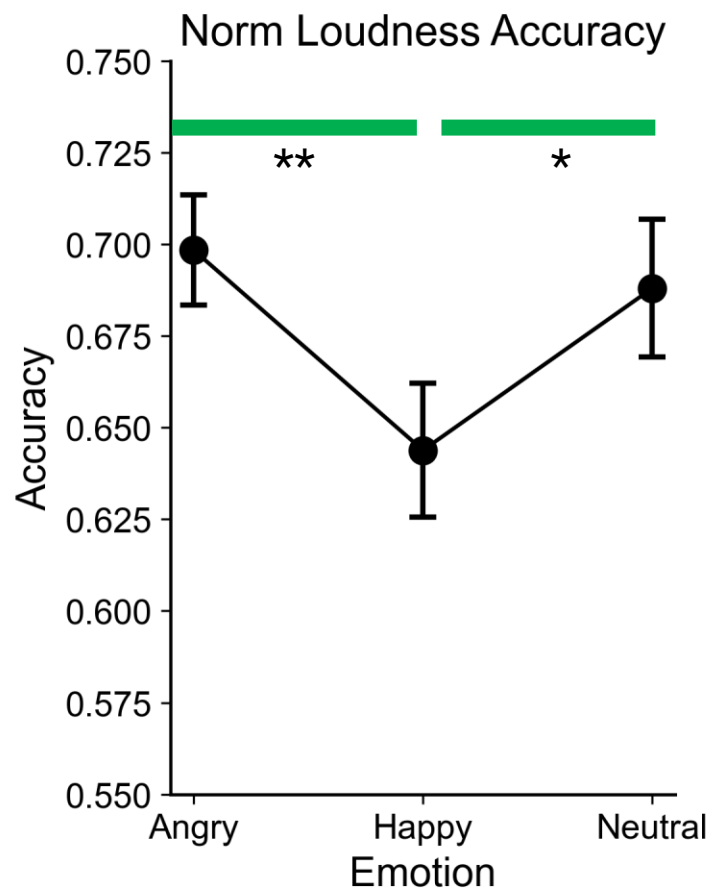
Perception of loudness modulation



- Accuracy varied across emotions ($F(2,104) = 3.388, p = .038, \eta^2_p = .061$)
- Loudness modulation for **happy prosody was less accurately identified** compared to angry ($p = 0.025, d = 0.4$).

Q

Loudness



Normal vs Quilted (Loudness)

Speech type:

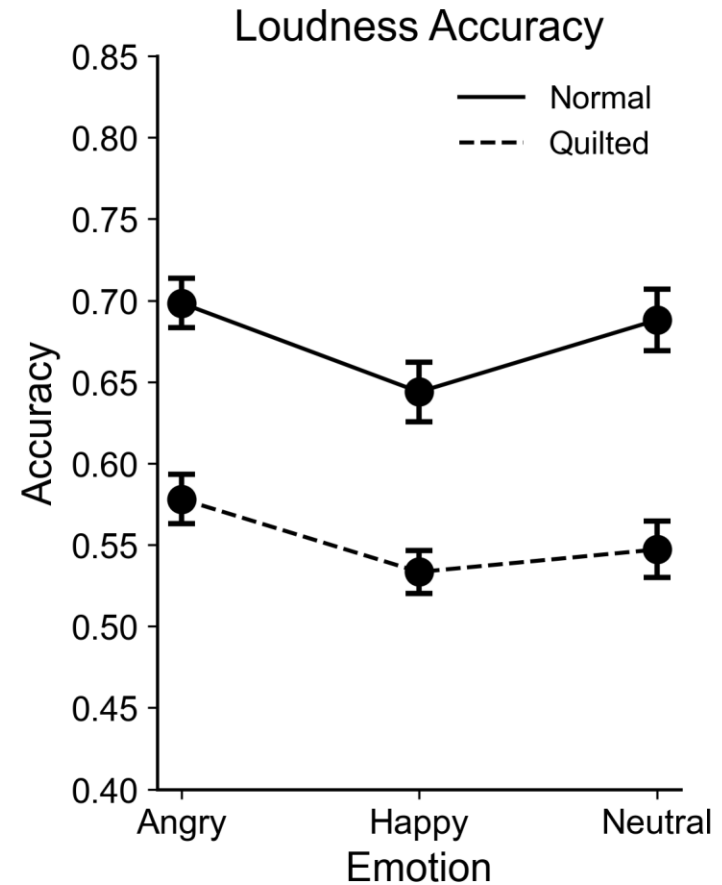
$F(1, 101) = 46.193, p < .001, \eta^2_p = .314$

Emotion:

$F(2,202) = 8.191, p < .001, \eta^2_p = .075$

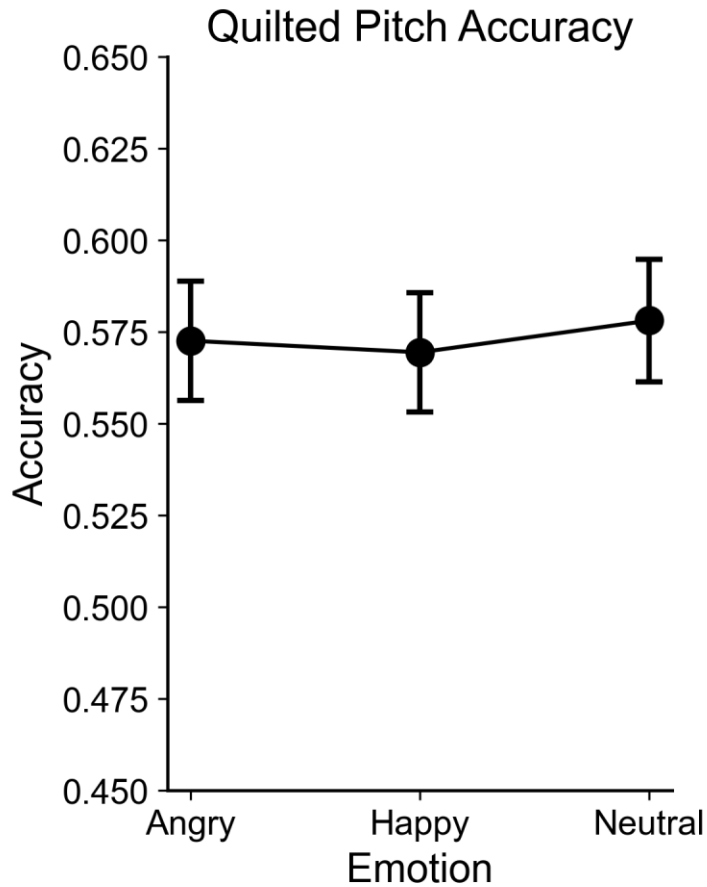
Interaction:

$F(2,202) = 0.791, p = .455, \eta^2_p = .008$



* $p < .05$
** $p < .01$
*** $p < .001$

Perception of pitch modulation



Accuracy did not vary across emotions
($F(2,100) = 0.109$, $p = .896$, $\eta^2_p = .002$)

Q

Normal vs Quilted (Pitch)

Speech type:

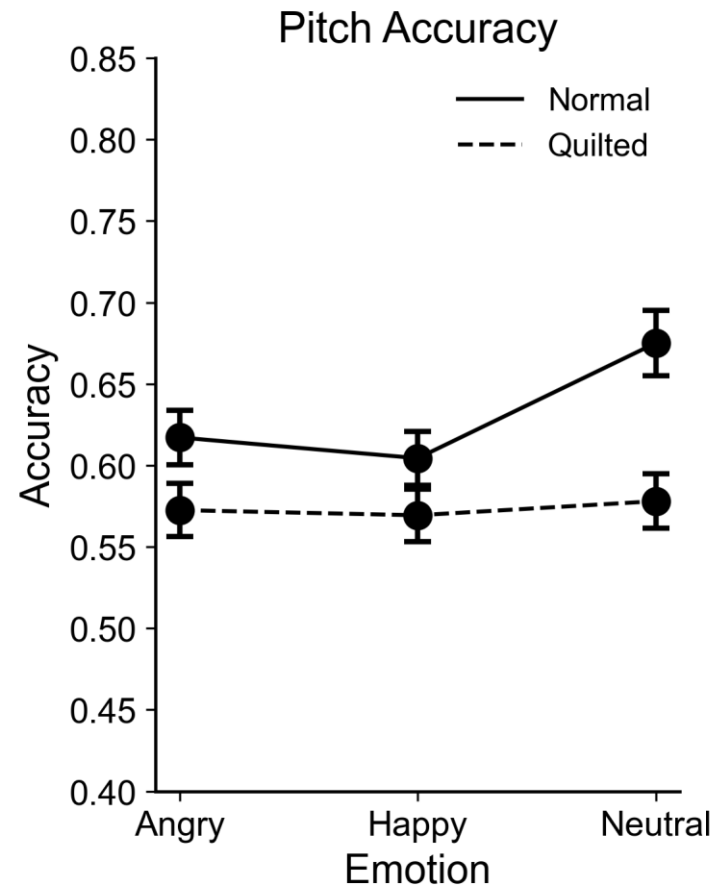
$F(1, 98) = 10.150, p = .002, \eta^2_p = .094$

Emotion:

$F(2,196) = 4.814, p = .009, \eta^2_p = .047$

Interaction:

$F(2,196) = 3.047, p = .050, \eta^2_p = .030$



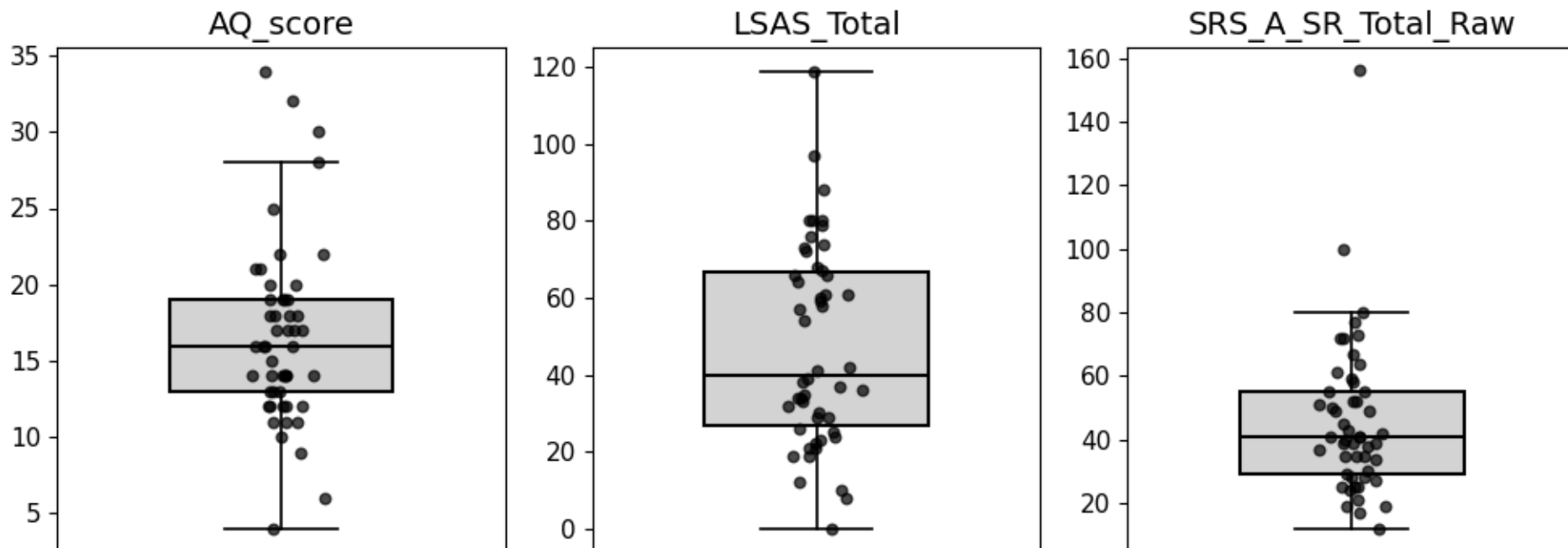
Results

Aim 2

Do Subclinical Psychopathology Traits Predict Differences in Accuracy Across Emotional Contexts?

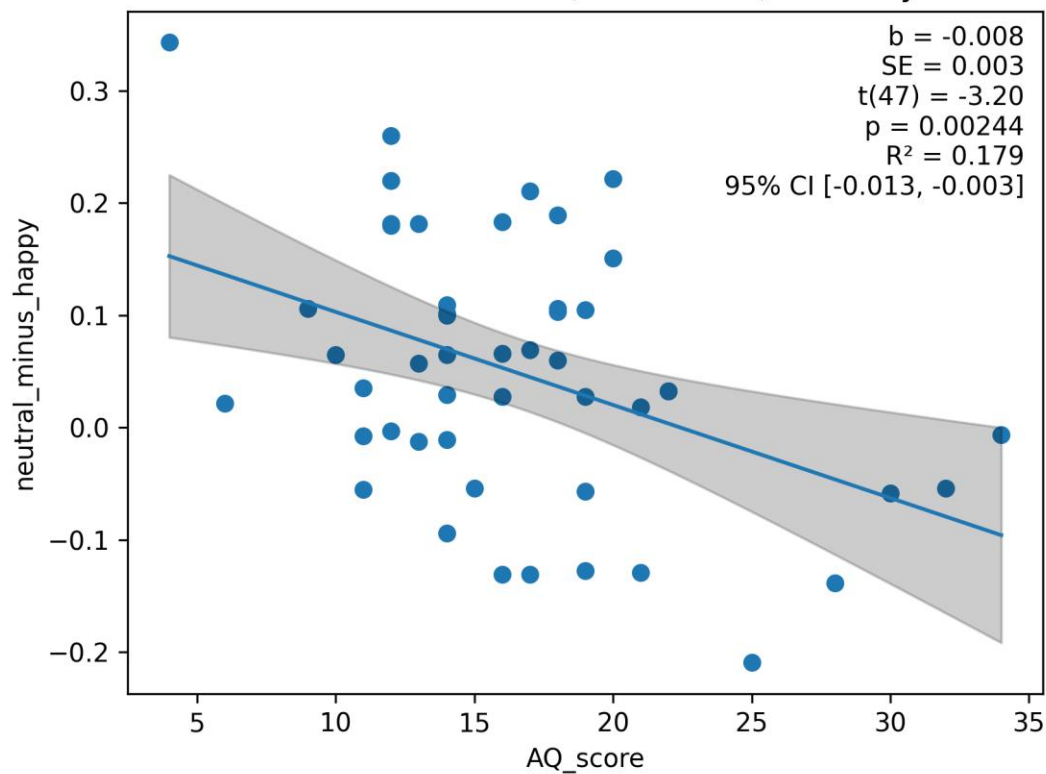
- Emotion difference scores
 - E.g., accuracy in happy – accuracy in neutral
- Used a linear model to see if psychopathology scores predicted performance

Measures of Psychopathology in "Typical" Sample

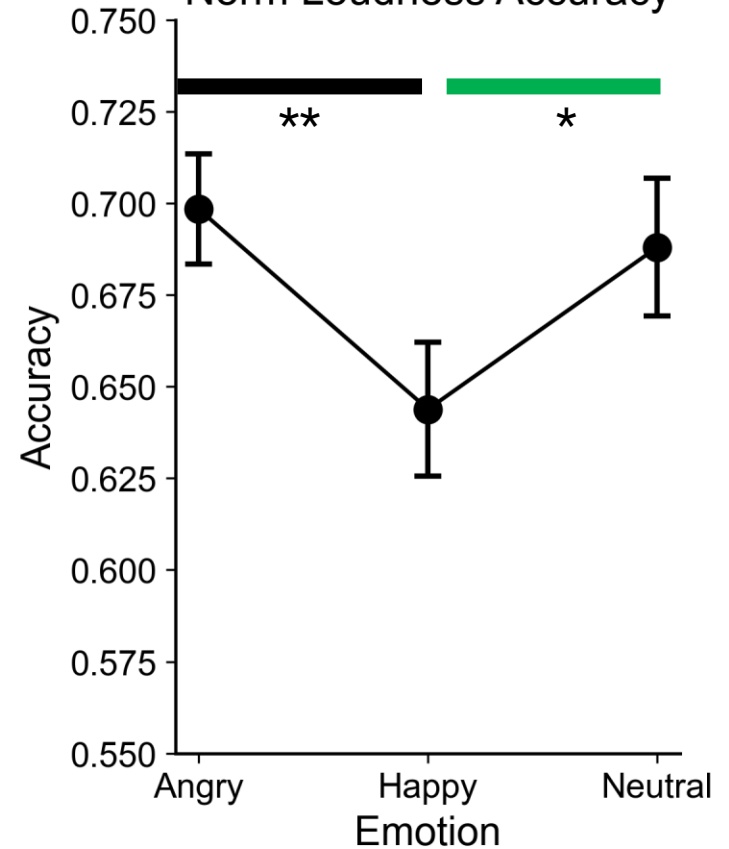


AQ

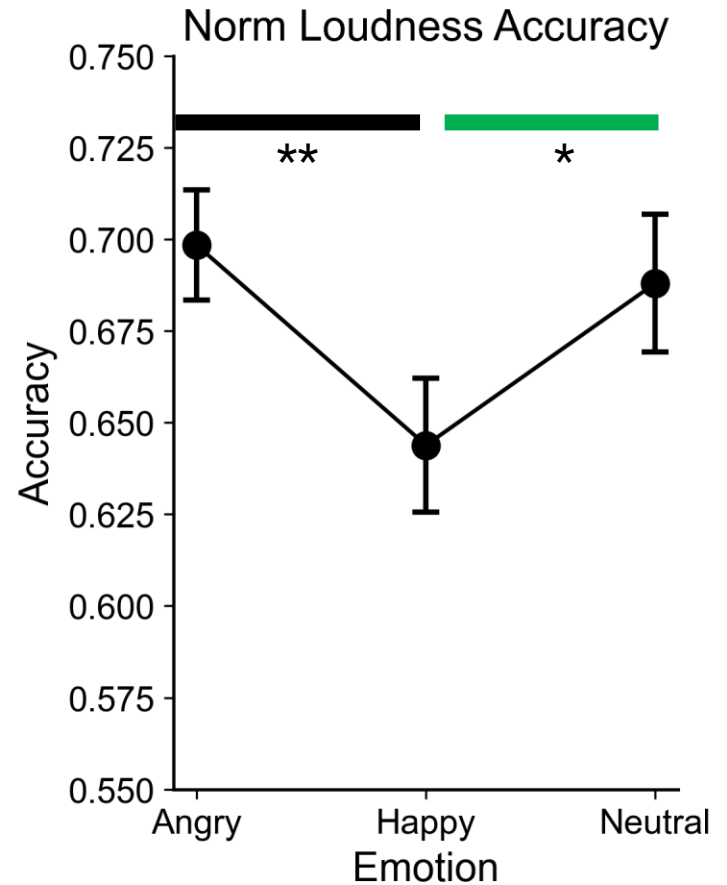
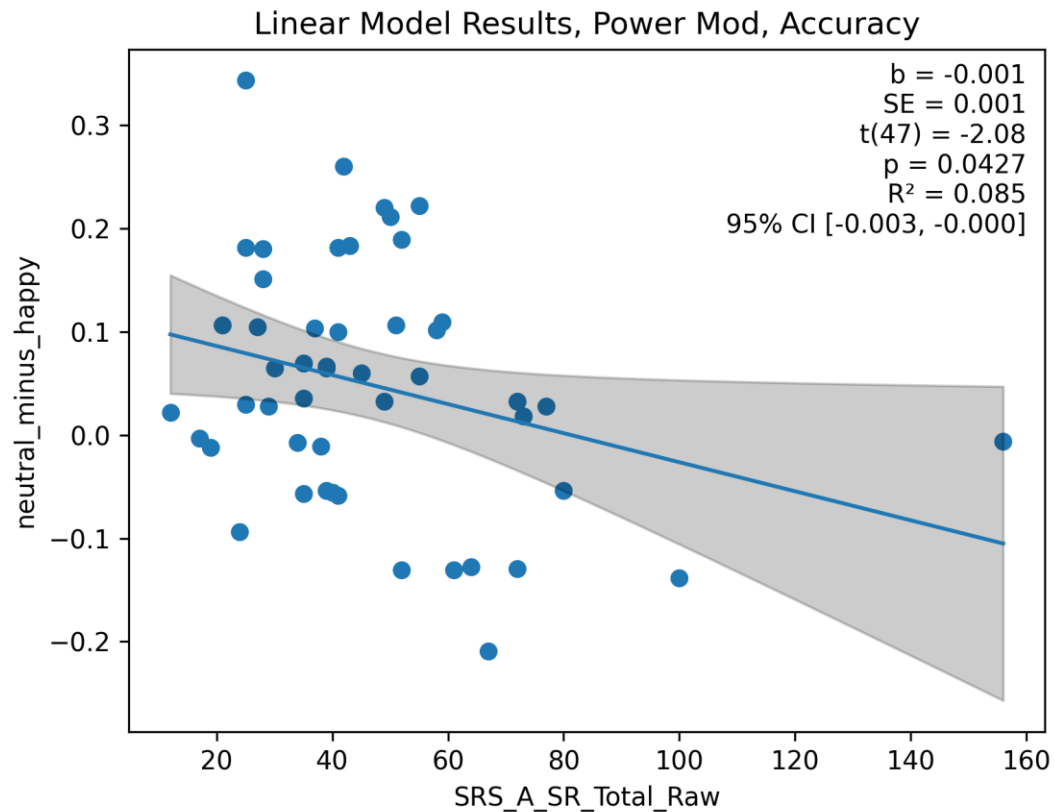
Linear Model Results, Power Mod, Accuracy



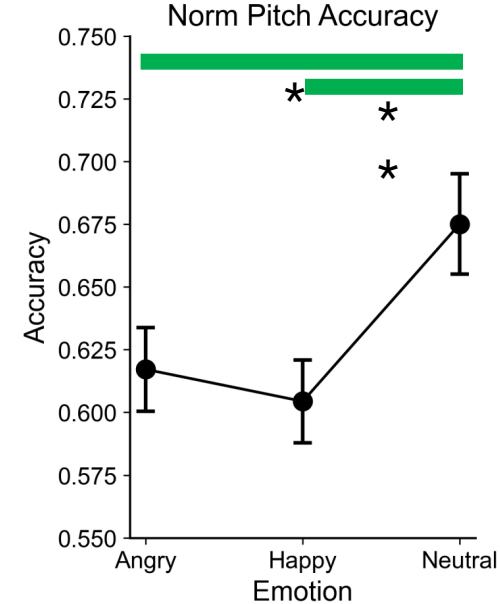
Norm Loudness Accuracy



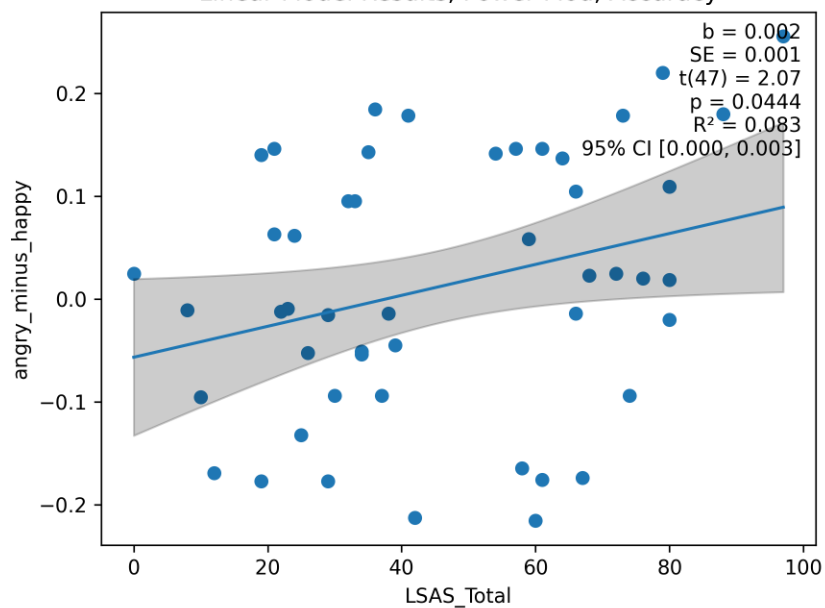
SRS Total



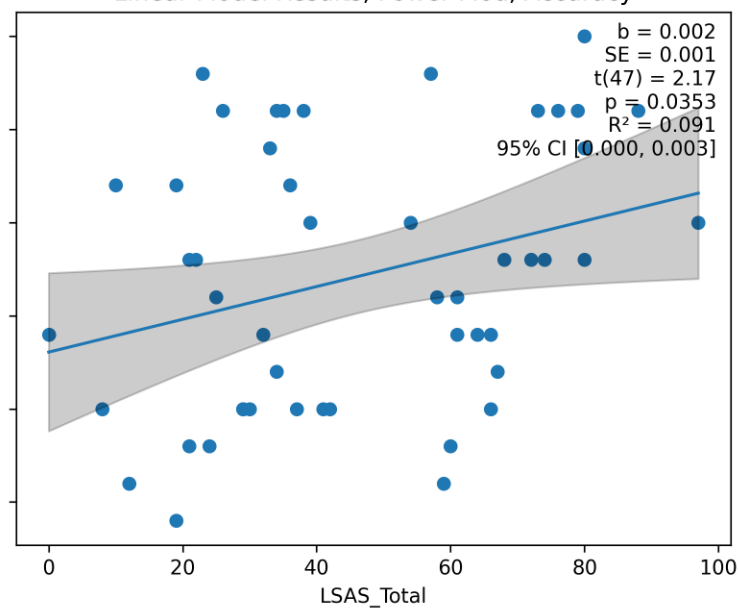
LSAS Total



Linear Model Results, Power Mod, Accuracy



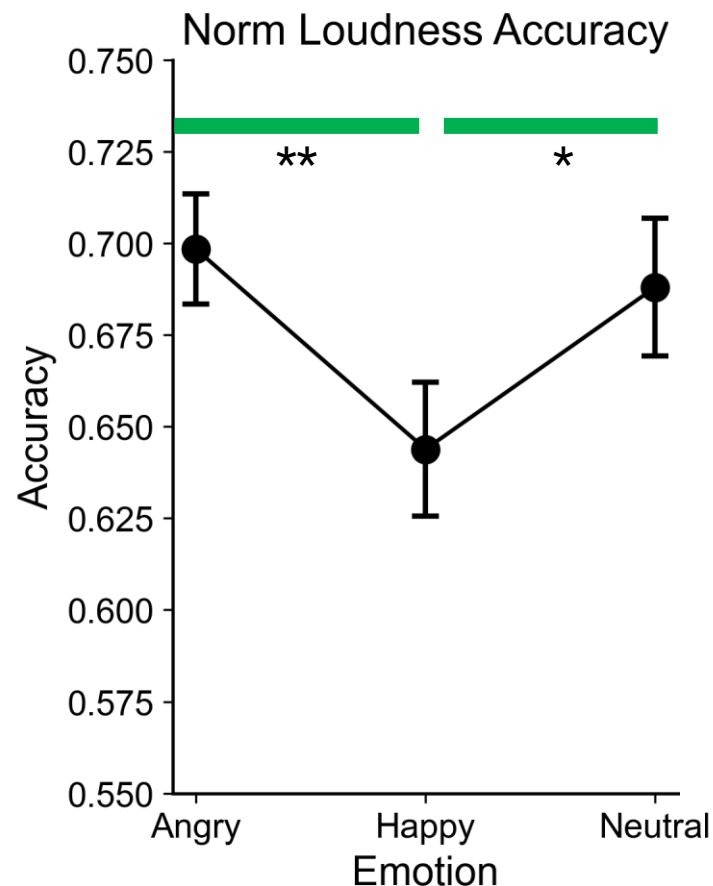
Linear Model Results, Power Mod, Accuracy



Discussion

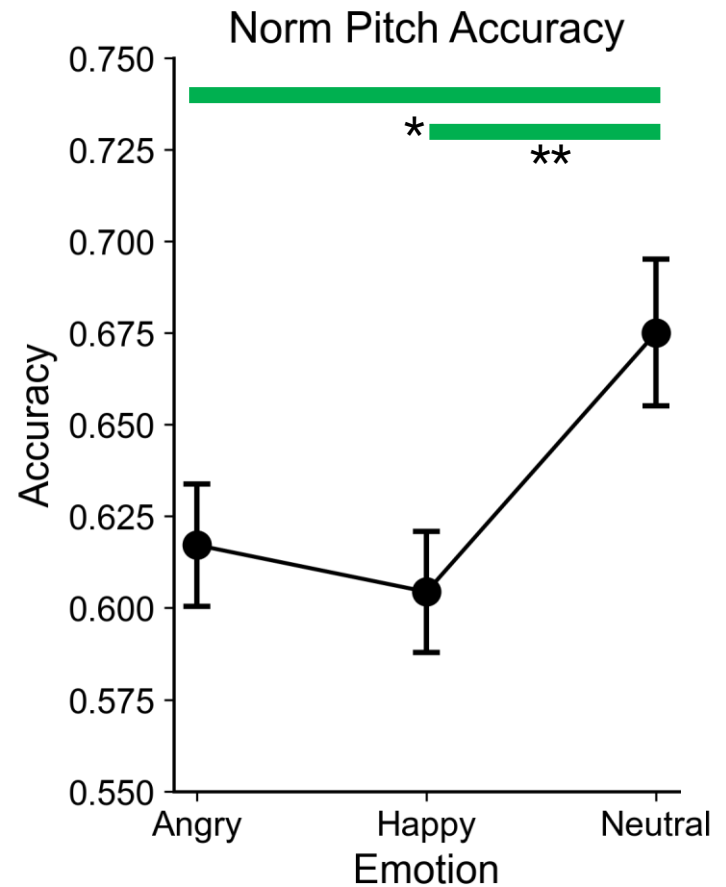
Different effects seen across modulations

- Emotional context influences low-level auditory feature perception
- Contextual dependence varies along psychoacoustic dimensions



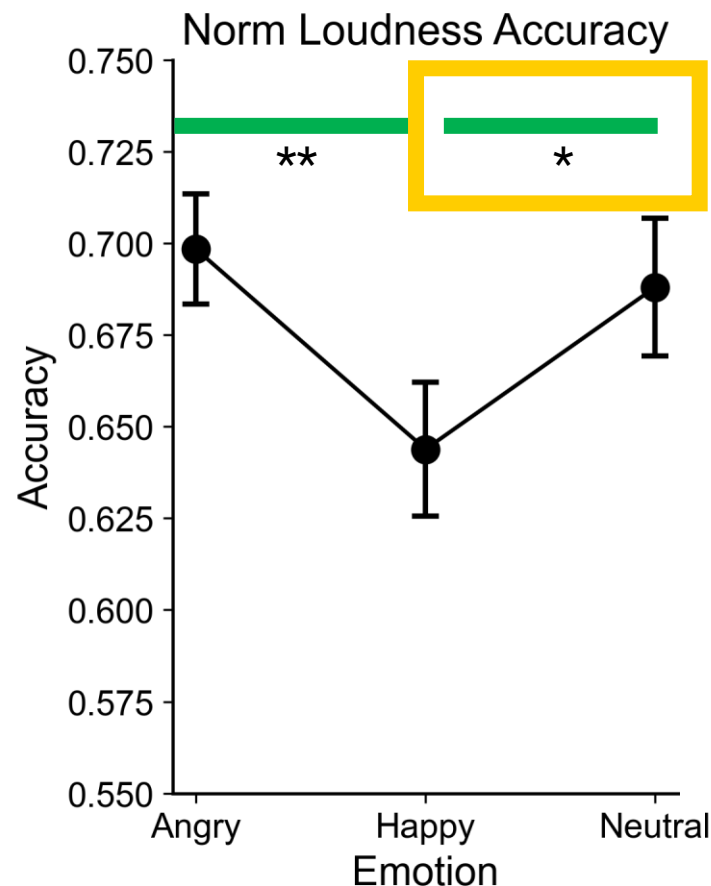
Different effects seen across modulations

- Pitch = emotion vs non-emotion
- Why?
 - Perceptual crowding?
 - Overweighting of prediction relative to sensory signal?



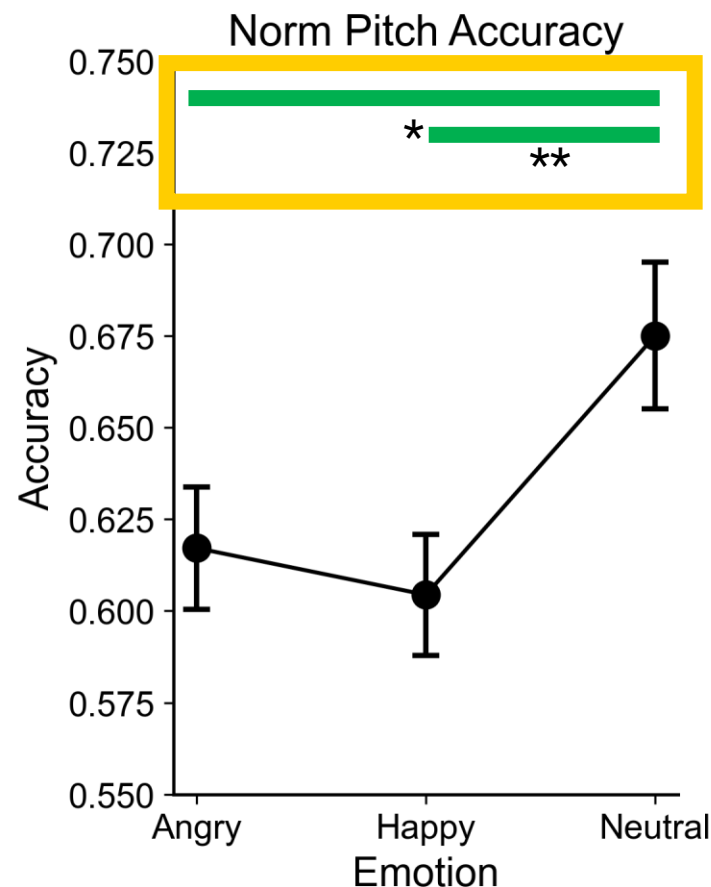
Psychopathology traits may contribute to observed effects (loudness)

- AQ and SRS Total Score are associated with happy vs neutral emotion difference
- No significant findings for social anxiety

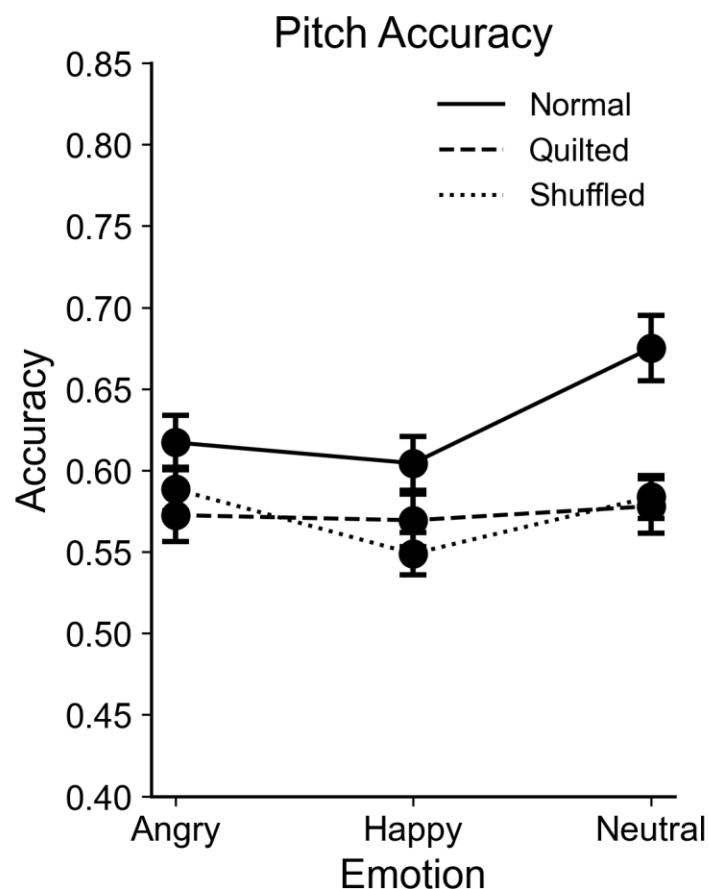
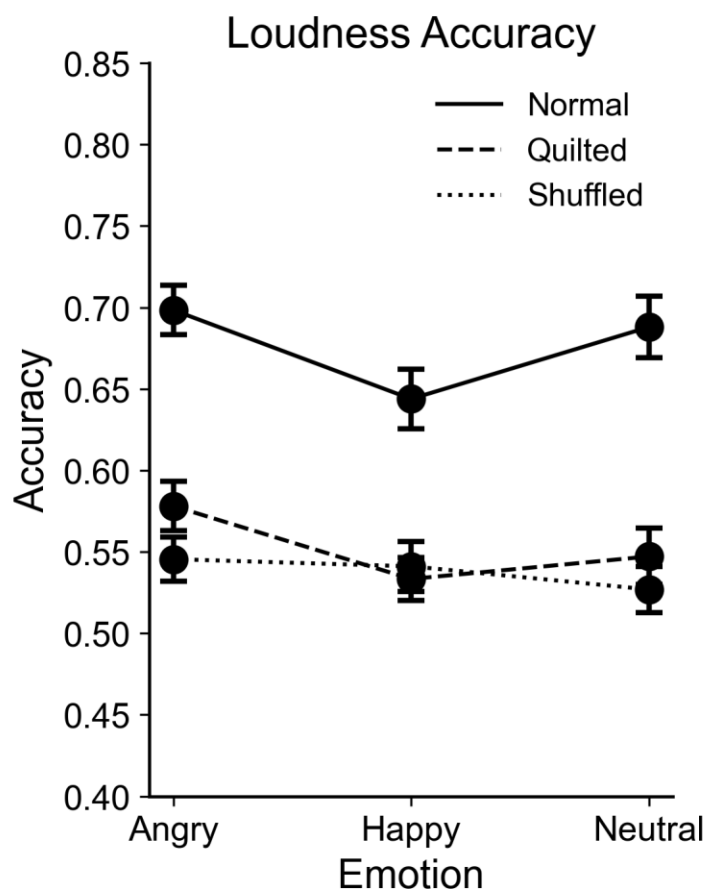


Psychopathology traits may contribute to observed effects (pitch)

- LSAS Total scores associated with both effects
- No significant findings for AQ or SRS



Prosodic context increases psychoacoustic sensitivity



THANK YOU!!

